



Original article

Guardian-AI: A novel deep learning based deepfake detection model in images

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ABSTRACT

The rapid advancement of deepfake technology has introduced significant challenges and opportunities across various domains. This article proposes a robust deepfake detection pipeline utilising a combination of attention mechanisms, pre-trained Vision Transformers (ViTs), and Long Short-Term Memory (LSTM) networks. The initial phase of the pipeline involves preparing photos and videos, potentially using optional facial detection to enhance accuracy. Vision Transformers derive features by capturing the global dependencies of input data. Long short-term memory (LSTM) networks address inter-frame temporal dependencies, whereas multi-head and traditional attention processes focus on essential elements. Ultimately, fully connected layers are employed for classification within the ensemble architecture, which consolidates the outcomes of several models. To ensure generalisability, assessment and regularisation approaches are employed to train the model using labelled datasets. Given the escalating threat of deepfakes, the findings indicate that the pipeline can consistently distinguish between genuine and fabricated information.

1. Introduction

Deepfake technology has revolutionised digital media manipulation. It facilitates the creation of synthetic content that closely resembles authentic media, rendering differentiation practically impossible [1,2]. Deepfakes can generate or alter material with remarkable realism due to the utilisation of advanced AI and ML algorithms. Although this technology holds significant potential for creative industries such as movies and entertainment, it presents substantial challenges for areas including politics, social media, and cybersecurity [4]. Researchers, lawmakers, and the public are apprehensive about the potential use of deepfakes for defamation, fraud and disinformation campaigns. Consequently, immediate implementation of efficient detection techniques is important [5].

Deepfakes possess the capacity to erode public confidence in digital

media and have surfaced as a potent instrument for disseminating deceptive information. Due to the accessibility of deepfake production tools, which facilitate the creation and distribution of disinformation by hostile actors, computer vision experts developed a facial re-enactment technique known as Face2Face [6]. This technique rapidly transforms an individual's facial emotions into a digital "avatar." Instances of deepfakes are being employed to manipulate political discourse, perpetrate financial fraud, or mimic prominent individuals [9].

In their general discussion of DeepFakes, the authors of [1] propose the R.E.A.L. framework as a means of managing DeepFake risks [10]. Attackers can now employ generative adversarial networks (GANs) to learn from image collections and create an image that looks realistic to human viewers thanks to advancements in machine learning and deep learning techniques [11].

Machine learning models trained to identify discrepancies or

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artefacts produced during the deepfake generation process are now employed for deepfake detection [2]. Convolutional Neural Networks (CNNs) have been widely utilised owing to their efficacy in executing image recognition tasks [12]. Several models, including ResNet, Xception, DenseNet, and Inception, have exhibited varying degrees of efficacy in detecting manipulated media [13]. However, novel deepfake generation algorithms exploit deficiencies in previous models to produce progressively more imperceptible data.

Additionally, in the case of individual video frames spatial features are considered as key parameter for numerous detection algorithms overlooking the parameter of temporal correlations that could contribute in terms of discriminative information. Considering Dynamism related to videos, efficient detection of finding gaps amongst successive frames could be augmented by temporal analysis. For the purpose of exhibiting sequential data Long Short-Term Memory (LSTM) networks has put forward technique for encompassing deepfake detection with temporal dynamics [14]. For researchers in computer vision; Vision Transformers (ViTs) and attention processes has caught hold of their eye in current times [15].

The capacity of models to recognise intricate patterns and overarching linkages is improved by focussing on the critical elements of the input data [12]. The interactions across various image regions are illustrated by Vision Transformers (ViTs), which have been successful in numerous image recognition applications [5,7]. Enlisted principles should be incorporated in deepfake detection which can contribute in tackling various shortcomings of traditional techniques based on CNN.

It is observed that there is dearth in techniques where varied latest models are integrated in order to enhance detection in terms of accuracy although some ongoing improvement can be sensed in detection methods [1,16]. Integration of various techniques comes with their own benefits which as if now has gone unnoticed in the field of research as still the focus remains on individual methodologies instead of ensemble methods [18]. Functionalities of various models are incorporated in the ensemble method that tends to address and deal with the issue of deepfake detection in a better manner [3,17]

This research introduces strong deepfake detection pipeline that integrate various advanced machine learning models to deal with the various concerns. The pipeline that is put forward is an inclusive system that evaluates video content by addressing its spatial and temporal characteristics. The proposed system comprises of pre-trained CNN, long short-term memory (LSTM), ViTs Vision Transformers and Attention Mechanisms [18]. The accuracy of detection is improved via pipeline deliberating on certain significant regions and optional face detection is included in the preprocessing phase [6,16].

ResNet, Xception, DenseNet, and Inception are some of the CNN architectures that helps in obtaining spatial features from frames. The models' efficacy in image analysis tasks and their capacity to learn hierarchical feature representations were the determining factors in their selection [9]. In contrast to convolutional neural networks (CNNs) [15, 19], vision transformers incorporate global dependencies. The pipeline employs long short-term memory (LSTM) networks to manage temporal relationships, which analyse frame sequences to identify outliers over time [20]. This temporal analysis is indispensable for identifying minute modifications that are imperceptible in individual frames but become apparent when examining frame sequences [14]. In order to enhance the model's capacity to differentiate between genuine and fabricated data, attention techniques such as multi-head and standard attention are employed to focus on critical features that earlier models were capable of extracting.

Outcomes of each of the models are combined in order to fit in their mainstream features; hence the ensemble architecture improves upon the operation of detection. To tackle with the insufficiencies of each of the models; the proposed methodology amalgamates their shared knowledge [21]. The input data is validated using the ensemble's architecture which is further tracked by the fully connected layers for the final classification's implementation [22]. Once the model is trained via

labelled data through supervision various distinguishing features of the deepfakes can be obtained. Although undergoing overfitting is the propensity of large neural networks, the challenge encountered above can be moderated using various evaluation and the regularisation techniques all through the training process by which the model performs excellently when introduced to the different data [23]. Primary findings of the pipeline's potential aids in distinguishing amid genuine and fake information which is the cutting-edge guard against deepfakes [24].

A new approach of incorporation of varied radical machine learning models to the context of integrated deep fake detection is been contributed by the undertaken research [16]. Classification and detection of deep fakes using attention mechanisms and taking into consideration the temporal and spatial characteristics of the video data is provided by this approach [8,10]. The findings of this research ensure reliability of media content and improve upon the digital security; further contributing to progressiveness of this domain and the future scope [4].

2. Related works

Advancement in Generative Adversarial Networks (GANs) has contributed immensely in the domain of deepfake creation, aiding in face alterations, modifications in expressions and identity transfer in the context of visual scenarios. In early times for deepfake detection CNNs were utilised but sensing temporal discrepancies remained an issue. Training of CNN architectures with huge datasets were established for enhancing the generalization across the video frames [25]. Modified and tailored face zones alongwith adjacent contextual extents have outpaced various customary classifiers by attaining advanced findings on standard datasets such as FaceForensics++, DFDC etc [16] for forgery detection with the help of identification inconsistencies in the Face- area.

Advanced disentanglement techniques reduce cross-forgery generalization and increase detection resilience across novel deepfake formats by separating forgery-agnostic features from domain-specific data [19]. Multi-modal transformers further improve identification by capturing multi-scale differences in both the spatial and frequency domains, which improves performance across a variety of datasets [12]. In oversampled datasets, augmentation techniques like Face-Cutout reduce the requirement for repeating samples while preserving important forgery traits by selectively obscuring non-critical portions [23]. In order to address highly compressed video situations, models such as FST-Matching are made to detect artifacts under compression, optimizing detection over a variety of quality levels [26]. Consistent assessment systems like DeepfakeBench presently offer standardized methodologies for reliable cross-model comparisons [21].

Temporal methods for cross-domain deepfake detection, such as ID-Reveal, rely on identity-based temporal disparities. This extends the model's applicability to unseen deepfake types and enhances detection without requiring labeled forgery data [14]. Knowledge distillation is used by transfer learning techniques like FReTAL to facilitate adaptive generalization in response to domain changes, ensuring continuity across domains without catastrophic forgetting.

For the purpose of domain-agnostic and reasonable feature extraction; modern approaches make use of disentanglement learning that tends to encourage reasonable detection around datasets which further removes preconceptions on the basis of demography [10]. With the aid of the process of Successive Subspace Learning(SSL) and feature extraction the DefakeHopmodel with 42,845 factors attains effective AUC scores when deal with large datasets that exhibits its efficacy and computational effectiveness [20].

With the optimization of the reliance of cross-quality features that empowers strong detection process amongst several datasets; the QAD i. e. Quality-Agnostic Deepfake advances detection across deepfakes of fluctuating nature [27]. By incorporating color-space pairwise connections through multi-channel Xception, the Multi-Channel Xception Attention Pairwise Interaction (MCX-API) model achieves comparable

accuracy on open-set standard datasets like FF++ and CelebDF, and establishes novel standards for generalised detection under various generative methods [17].

3. Methodology

In the proposed research work a system with all-inclusive solution for the deepfake detection is presented that makes use of contemporary machine learning models for improving the accuracy of detection and its robustness [3,6,7]. The four main constructs of the proposed system are data preprocessing, feature extraction, temporal modeling with attention mechanisms, an ensemble model architecture [15,28]. Fig. 1 explains the flow of the Guardian AI deepfake detection pipeline.

3.1. Data preprocessing

Image Preprocessing: The initial stage involves loading images and resizing them to a standard dimension of 224×224 pixels in order to ensure consistency across the dataset [23]. An extra face detection process using the Multi-task Cascaded Convolutional Networks (MTCNN) method can be used to focus the research on facial areas that are more susceptible to manipulation in deepfakes [8,13]. In order for the neural networks to train more effectively, the images are normalized to maintain consistent pixel value distributions during scaling and facial recognition [29], the whole process as being detailed in the Fig. 2.

3.2. Feature extraction

Vision Transformer (ViT) Extraction: In addition to CNNs, the system

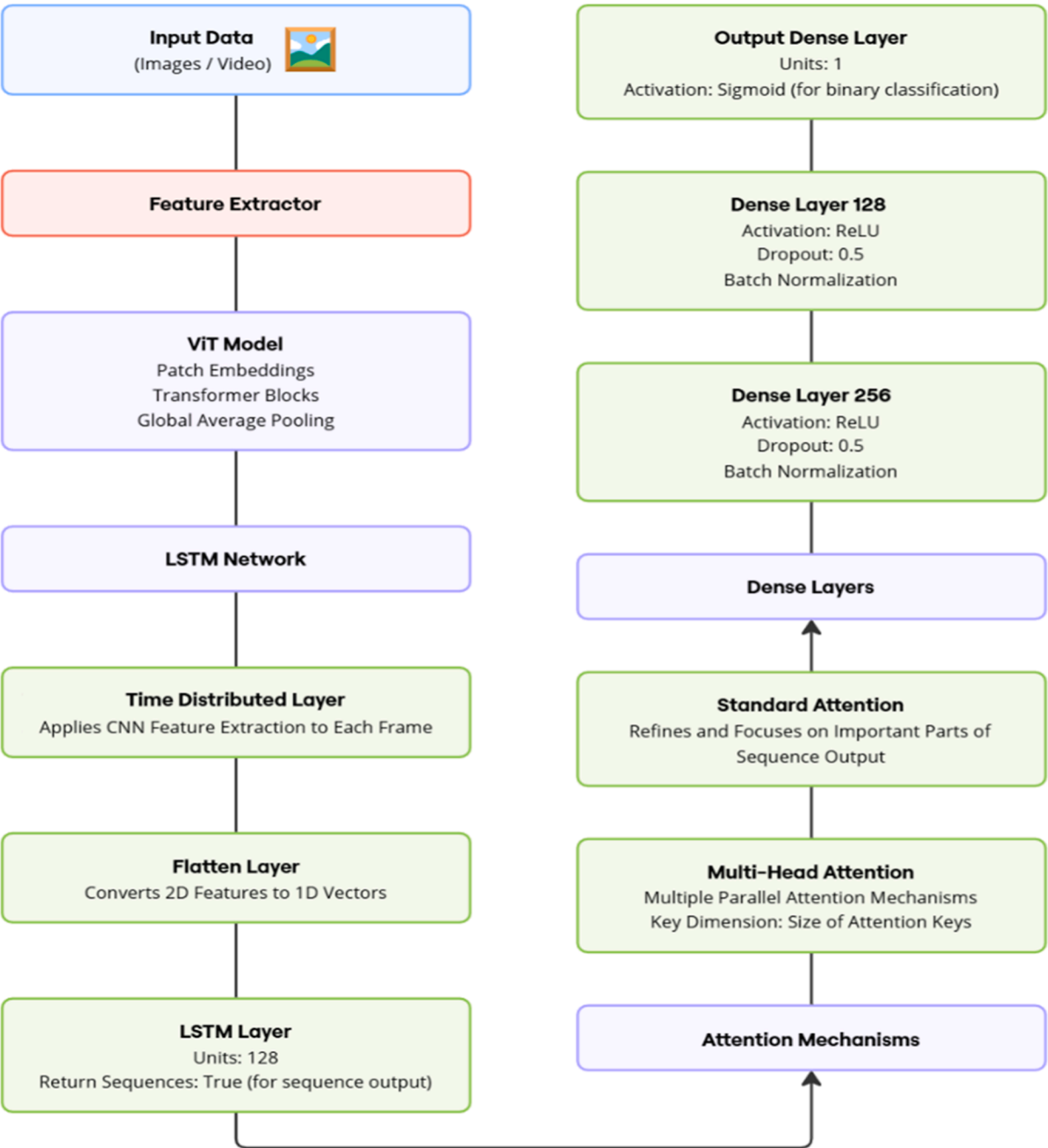


Fig. 1. Flowchart of the Guardian AI Deepfake Detection Pipeline.

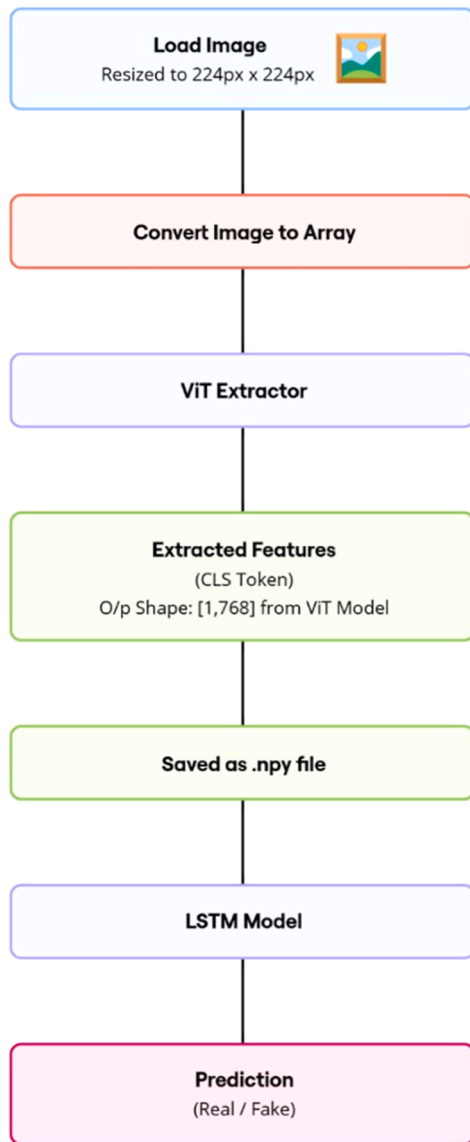


Fig. 2. Data flow diagram of Guardian AI deepfake detection pipeline.

uses Vision Transformers (ViTs) to extract global features from the input data [15]. By employing self-attention approaches to describe long-range dependencies in the images, ViTs offer an alternative perspective to traditional CNNs. This allows the model to identify intricate changes by identifying connections between far-flung areas of an image [10,28]. Usage and method involved in ViT for deepfake detection is shown in Fig. 3, whereas the ViT architecture is shown by Fig. 4.

3.3. Temporal modeling and attention mechanisms

LSTM Network: In the case of deepfakes related to videos; significance of temporal dynamics is spotted using Long Short-Term Memory (LSTM) network which is exploited to model interframe relationships in sequential fashion for video frames [14]. It is tedious to find temporal difference when frames are spotted individually whereas LSTM examines series of attributes sequentially to recognize temporal differences from successive frames. Fig. 5 depicts the architecture of LSTM.

Multi-Head Attention: Incorporating a multi-head attention mechanism to the outputs obtained by LSTM network additionally enhances temporal modelling. Spotting of significant temporal features advances the models ability to classify between genuine and fake matter also keeping a focus on several order of segments.

Standard Attention Block: To get an insight of the important facets related to input data assistance of the standard attention block is needed. Over the dimensions of time and space the attention block assigns weights to the various attributes that in turn makes the model more responsive to the substantial markers of deepfakes [7,26].

3.4. Ensemble model architecture

The essence of the Guardian AI approach is the ensemble architecture, which integrates the outputs from various components of the model and the feature extraction process to arrive at the ultimate classification decision [22]. The Vision Transformer's retrieved features are amalgamated to produce a feature vector that encapsulates many interpretations of the input data [17]. The model's ability to discern complex patterns and relationships is enhanced by processing this aggregated feature vector through the LSTM network, followed by multi-head attention and standard attention mechanisms. The outputs are transmitted to a series of fully connected dense layers. To enhance generalisation and mitigate overfitting, they employ batch normalisation and dropout approaches [11,19].

The final classification layer evaluates the legitimacy of the input media by employing these processed attributes [2]. The ensemble design is a dependable system that can precisely identify deepfakes by

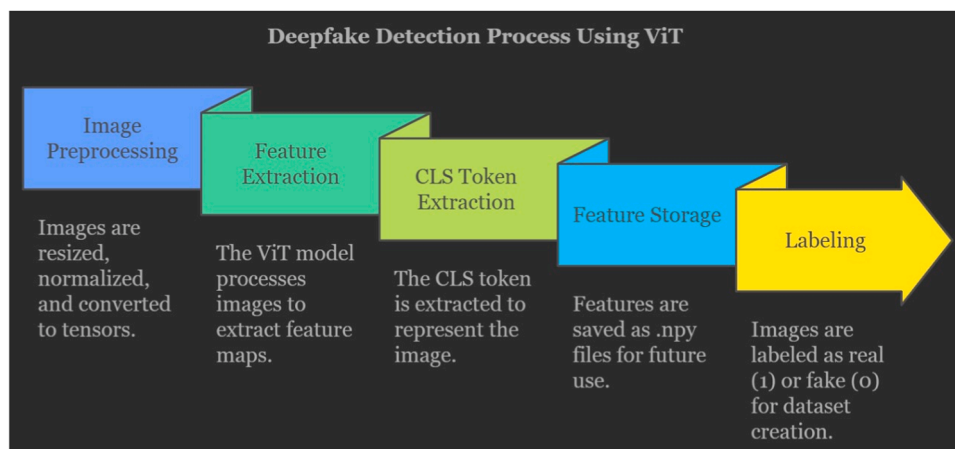


Fig. 3. Deepfake detection process using ViT.

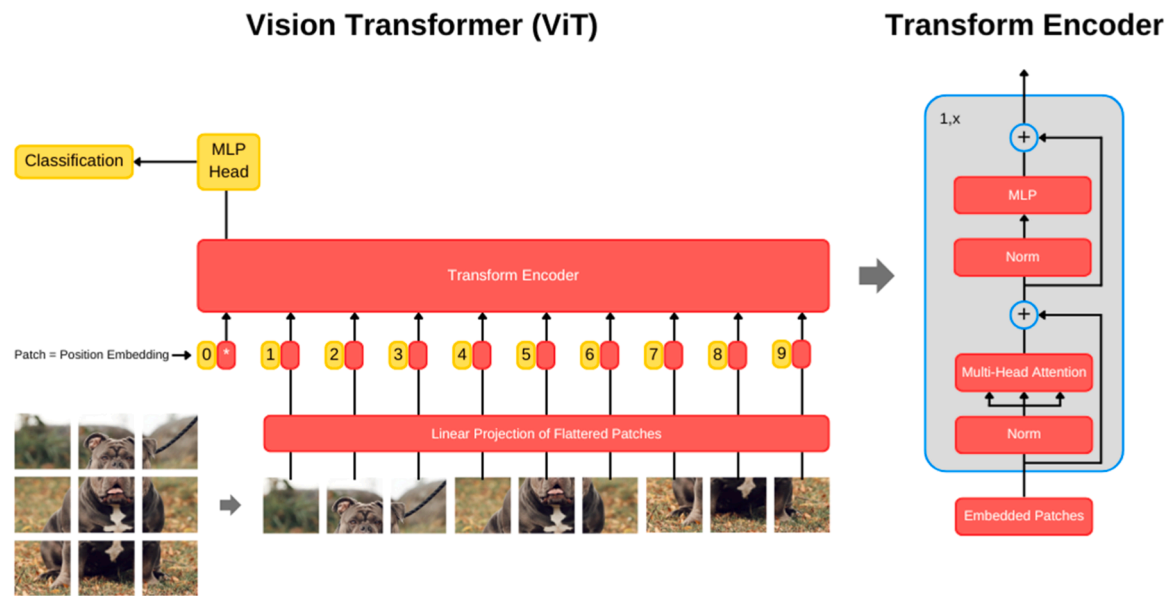


Fig. 4. ViT (Vision Transformer) architecture.

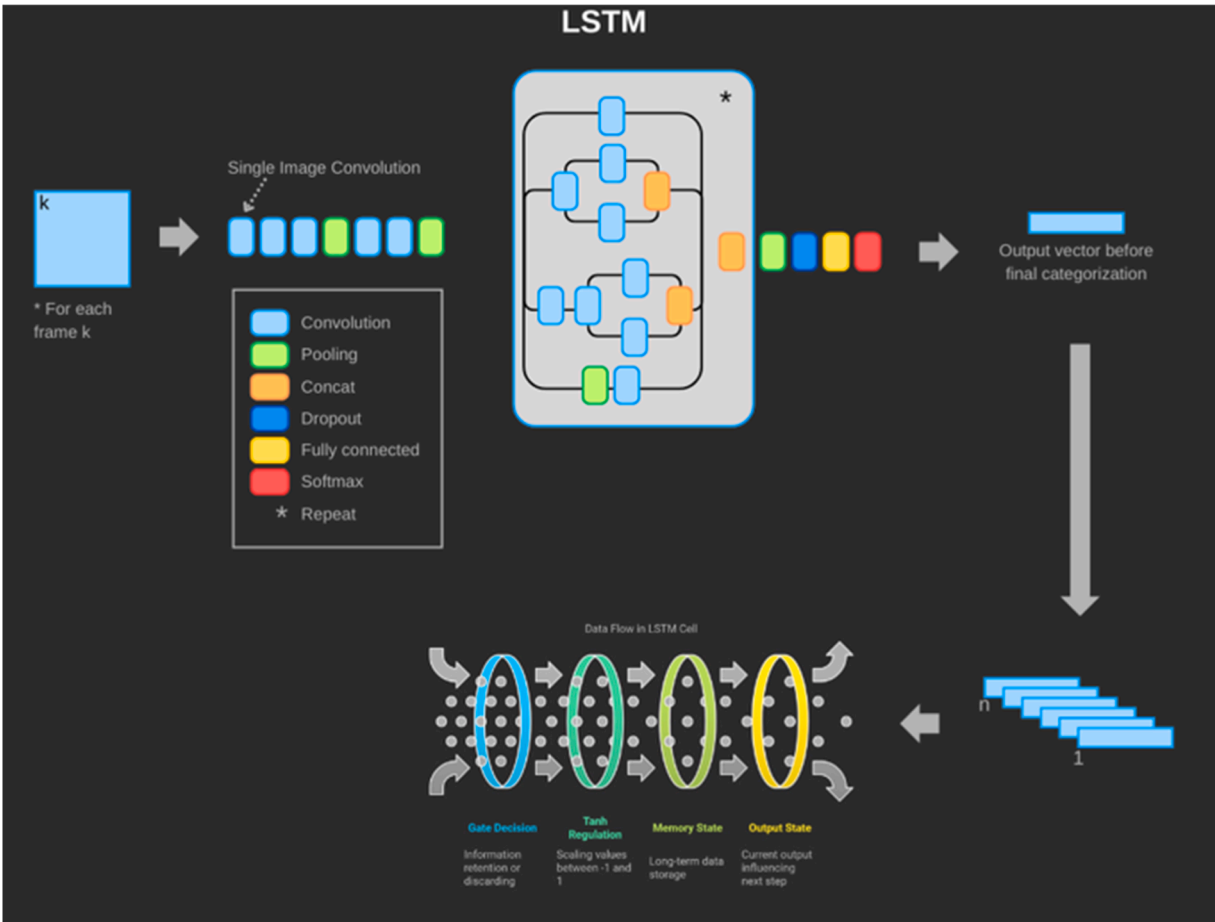


Fig. 5. LSTM (Long Short-Term Memory) architecture.

integrating many models and approaches, leveraging the strengths of each component [16,23,25,28].

4. Results

To assess the performance of the proposed Guardian AI approach; it is applied on CDDDB (own Deepfake Detection Benchmark) dataset which deals with complex deepfake scenarios in the real-world. Guardian AI

demonstrated its proficiency in managing a wide range of complex and high-quality facial alterations, achieving a 95.8 % accuracy rate on CDDB. The exceptional accuracy results from the ensemble design's i.e. incorporation of Vision Transformers (ViTs), Long Short-Term Memory (LSTM) networks and multi-head attention mechanisms, which address spatial and temporal irregularities in deepfake data. Variation in accuracy computed Fig. 6 illustrates how accuracy varies when applied to benchmark datasets [10,23,25,28].

A low false-positive rate, evidenced by a precision of 96.2 %, restricts the misclassification of authentic photos as deepfakes. The performance of Guardian AI on CDDB was assessed using these significant metrics of performance.

- Even in the most challenging settings with minor alterations, Guardian AI was able to determine various CDDB manipulation tactics with a recall rate of 95.6 %.

- The model exhibits uniform effectiveness across multiple CDDB samples, evidenced by an F1-score of 95.9 %, indicating a balanced detection capability. Fig. 7 illustrates the training accuracy of the deepfake detection pipeline, while Fig. 8 depicts the validation loss.

Guardian AI successfully handled a variety of deepfake difficulties seen in the CDDB dataset, including identity swapping, facial reenactment, and emotion modifications. Global dependency modeling was made possible by the incorporation of Vision Transformers, which helped to capture subtle facial irregularities that traditional CNNs could overlook. Guardian AI's sensitivity to frame-to-frame inconsistencies—a crucial component in video-based deepfake detection—was further improved by the multi-head attention mechanism in conjunction with LSTM networks for temporal analysis.

The usefulness of integrating model strengths was demonstrated by the 5 % improvement in overall accuracy that the ensemble approach on CDDB produced when compared to single-model baselines. The CDDB exemplifies the application of Guardian AI in critical, high-stakes scenarios where accuracy is paramount, owing to its balanced precision and recall. However, in high-compression settings, performance was slightly reduced, suggesting that future advances could revolve upon optimising

input data in real-time for different quality standards.

Other deepfake detection techniques such as custom CNN and DenseNET architecture has also been implemented using the CelebA+FFHQ and face forensic benchmark datasets [31]. In the Custom CNN architecture each of the six convolutional layers (Conv2D) in the Custom CNN architecture was paired with a max pooling, dropout, and batch normalisation layer. The input layer used the Rectified Linear Unit (ReLU) activation function, whereas the output layer made use of the sigmoid activation function. In DenseNET architecture; each layer had its own dropout to prevent over-fitting, and the kernel had its own padding to improve image accuracy. This study makes advantage of Keras's DenseNet-264 architecture, which adds a dense layer as the final output layer. The architecture begins with a convolutional layer with 7×7 stride 2 and is followed by a MaxPooling layer with 3×3 stride-2. The output layer uses a sigmoid activation function, whereas the four dense blocks on the input layers use batch normalisation and the ReLU activation function. Comparative analysis of Guardian AI has been performed with other deepfake detection techniques such as Custom CNN architecture and DenseNET Architecture in terms of Accuracy as shown in Fig. 9. Guardian AI outperformed other deepfake detection techniques.

5. Conclusion

The newly suggested deepfake detection system, Guardian AI, incorporates multiple deep learning models, including LSTM networks, Vision Transformers (ViTs) and attention mechanisms, among others. The ensemble method, created by Guardian AI, utilises the advantages of several models to enhance detection precision and robustness against deepfakes. The system's emphasis on essential components is significantly enhanced by the use of Multi-Head Attention and standard attention blocks. Guardian AI is a significant advancement in deepfake detection, providing an effective approach in maintaining the authenticity of multimedia content. Future study will focus on the application of this framework for diverse media manipulation, the integration and

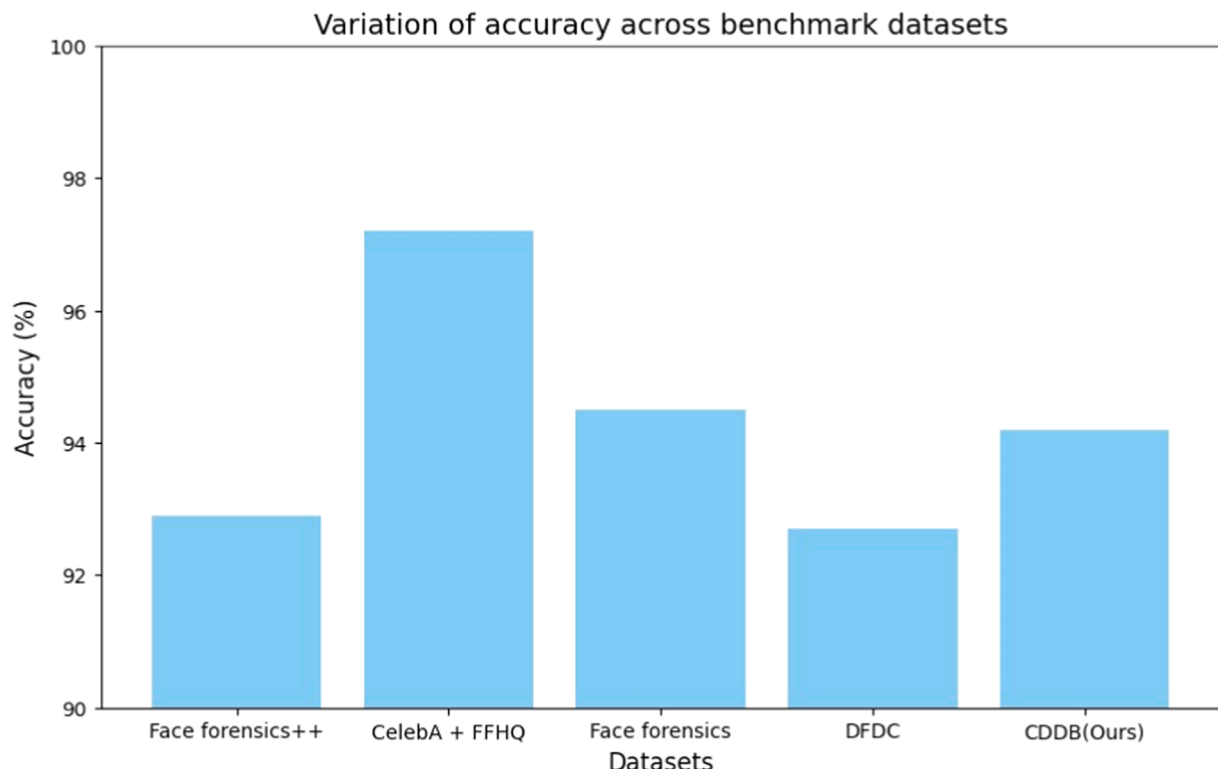


Fig. 6. Accuracy of guardian AI Across benchmark datasets.

Training Accuracy

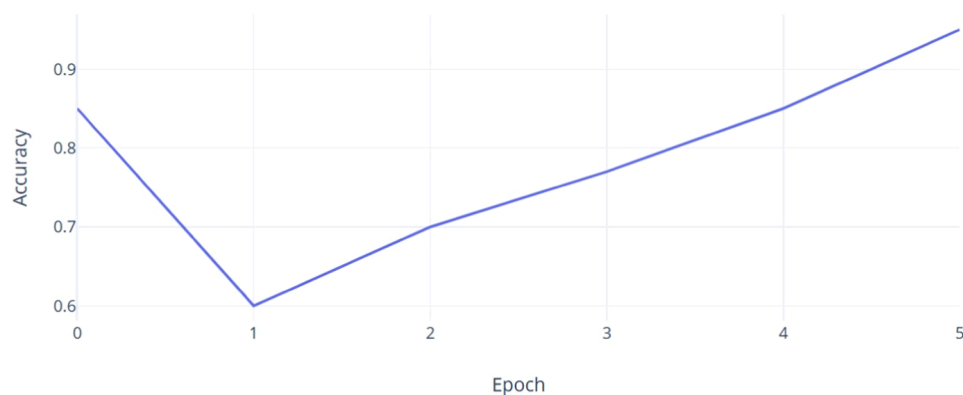


Fig. 7. Representation of Training Accuracy.

Validation Loss

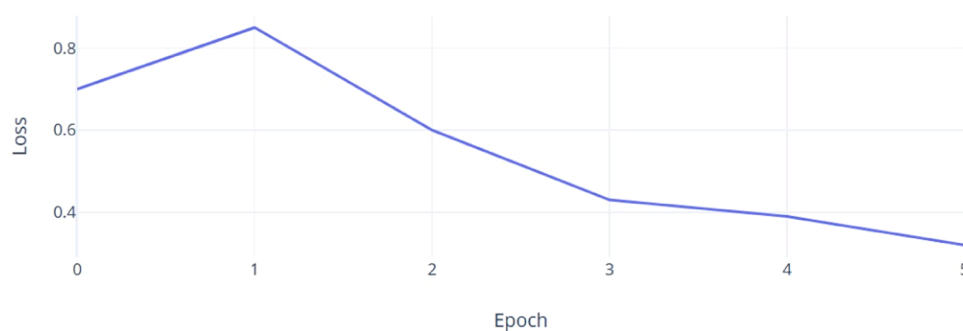


Fig. 8. Representation of Validation Loss.

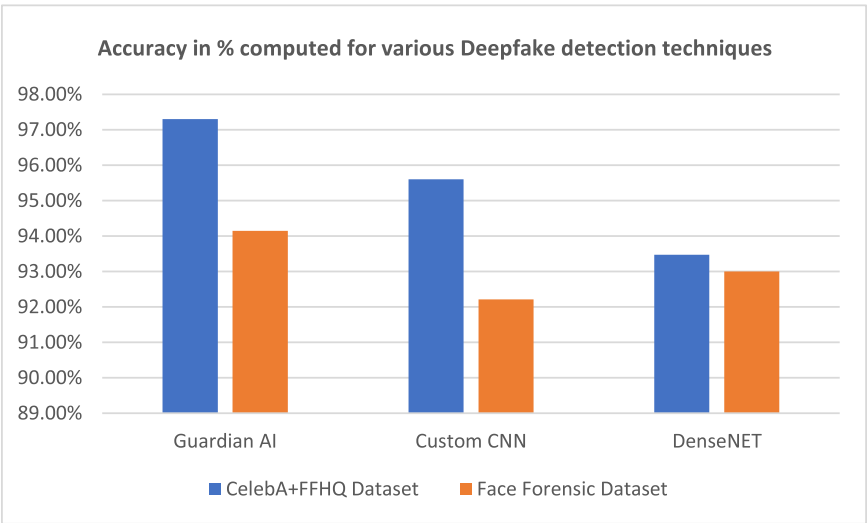


Fig. 9. Accuracy in percentage for various deepfake detection techniques for various benchmark datasets.

incorporation of additional models and the ability for real-time detection. When compared to other deepfake detection approaches, Guardian AI fared better.

Key findings

Many different types of images and videos, both real and deepfake, were used to assess the Guardian AI. The ensemble model outperformed single model tactics in detecting deepfakes, achieving high accuracy.

With the use of LSTM networks, attention mechanisms, and ViTs; Guardian AI successfully detected anomalies in both space and time. Accuracy, precision, recall and F1-score were some of the metrics used to assess the model's performance; these measures showed how effectively the system distinguished between modified and original content.

Future scope

In order to make the Guardian AI framework more useful for surveillance and live streaming scenarios, future research could concentrate on improving its ability to identify deepfakes in real-time [9,10,30]. Combining adversarial training methods with sophisticated generative models may improve resilience against developing deepfake techniques [5,22]. Broadening the method's applicability to identify audio deepfakes and multimodal manipulations would further mitigate the risks associated with synthetic media [3,4,24]. The model's applicability to a wide variety of contexts and technologies can be enhanced even more by collaborative efforts on bigger and more varied datasets [17,21].

CRediT authorship contribution statement

Alqahtani Mohammed: Writing – review & editing, Visualization, Validation, Software, Data curation. **Ben Miled Achraf:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology. **Alshuhail Asma:** Writing – review & editing, Writing – original draft, Validation, Resources, Conceptualization. **Mahmood Khalid:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Alghamdi Bandar M.:** Writing – review & editing, Writing – original draft, Methodology. **Alallah Fouad Shoie:** Writing – review & editing, Writing – original draft, Validation. **Alshareef Abdulrhman:** Writing – review & editing, Writing – original draft, Validation. **Alsolai Hadeel:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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